Parent Guide — middle / module-06-moon-phases-tithi

IKS Curriculum

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Parent Guide — Module 6: Moon Phases & Tithi

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min over ~2 weeks

What your child is learning

The traditional Indian lunar calendar — the *pañcāṅga* — and how it tracks the changing shape of the Moon through a unit called the *tithi*:

Sanskrit	English
tithi	a lunar "day" — 1/30 of a full Moon cycle
pakṣa	a fortnight (half of a lunar month)
śukla pakṣa	the waxing fortnight (Moon growing)
kṛṣṇa pakṣa	the waning fortnight (Moon shrinking)
amāvasyā	new moon
pūrṇimā	full moon
pañcāṅga	the five-limbed almanac (tithi + 4 others)

Plus a deeper layer: - **Why the Moon changes shape** — the Sun lights up only half of the Moon, and we see different fractions of that lit half as the Moon orbits Earth. - **Why eclipses happen** — when Sun, Earth, and Moon line up exactly, and only at new moon (*amāvasyā* — solar eclipse) or full moon (*pūrṇimā* — lunar eclipse). - **Why a tithi is not a day** — because the Moon doesn't move uniformly. A tithi can last anywhere from ~21.5 to ~26 hours.

How we teach it

The framing is **secular and scientific first**. We lead with what your child can observe in the night sky, then situate the *pañcāṅga* as the way Indian astronomers organised those observations over many centuries. We do NOT teach this instead of modern astronomy — we teach both side by side.

NEP 2020 §4.6 (Integration of Indian Knowledge Systems) supports this approach.

Homework you'll see

- **Nightly Moon-watching diary** — 10 nights of looking up at the Moon, drawing its shape, noting the tithi (Days 1–10).
- A 1-month tithi calendar your child builds for the current month (Day 4).
- A short discussion on lunar months and festivals (Day 5).
- A project — poster, slideshow, or 90-second video on one lunar phenomenon (Days 8–10).
- 6–8 writing prompts across the module.

How to help

Helpful: - Step outside with your child for 5 minutes one night and look at the Moon together. Ask what shape it is. That's the whole assignment. - If clouds block the sky, look at a Moon-phase app or NASA's published phase calendar with them. - For the tithi calendar (Day 4), help them locate any standard *pañcāṅga* — a printed almanac, a temple calendar, or an app like Drik Panchang or Mypanchang. Use the same one consistently. - Talk about festivals in your family that are tied to specific tithis — Saṅkrānti, Dīpāvalī (amāvasyā), Holī (pūrṇimā), Karva Cauth (caturthī of Kṛṣṇa pakṣa in Kārtika), Janmāṣṭamī (aṣṭamī of Kṛṣṇa pakṣa in Bhādrapada), etc. Real local examples make the system stick.

Not helpful: - Insisting on a particular religious interpretation of any tithi or eclipse. The classroom keeps the astronomical content primary. - Doing the calendar for them. The lookup is the learning. - Pushing your child toward a "correct" position on whether to follow *pañcāṅga* timings personally. That's a family matter.

Safety — eclipse viewing

If a solar eclipse occurs during this module, **never let your child look at it directly** without ISO 12312-2 certified eclipse glasses. Ordinary sunglasses do not protect the eyes. The safest method at home is pinhole projection onto a sheet of paper. Lunar eclipses, by contrast, are safe to look at directly.

We cover this in Day 7. Reinforce it at home.

Citation standard

By Middle band, projects should reference at least one source — either a modern astronomical reference (NCERT Physics XI, NASA's phase calendar) or a paraphrase of the *Sūrya-siddhānta* (Indian astronomical text, compiled c. 5th–10th c. CE). We do not quote specific verses at Middle band because verifying a single verse number requires scholarly editing; we paraphrase instead and cite the text by name.

What success looks like

By end of module, your child can: - Explain in their own words why the Moon changes shape. - Name the two pakṣas and at least the four "anchor" tithis (*pratipadā*, *aṣṭamī*, *pūrṇimā*, *amāvasyā*). - Read a tithi from a *pañcāṅga* for any given date. - Explain why eclipses happen and why they don’t happen every month. - Present a 3-minute project on one lunar topic with a clear source citation.

On respect across traditions

The *pañcāṅga* is used both for everyday timing (planting, travel) and for religious observance (fasts, festivals). The classroom honours both uses without privileging either. If your child brings home questions about a specific tithi’s religious significance, that’s a conversation for your family — the school covers the calendrical and astronomical layer.

Want to go deeper?

Senior band (Grade 9–12) covers the actual *Sūrya-siddhānta* algorithm for tithi computation, compares the Indian system with the Babylonian and Hellenistic traditions, and discusses *adhika māsa* (intercalary month) calculation. Preview: `curriculum/senior/module-06-moon-phases-tithi/README.md`.

10-Day Lesson Plan

Module 6 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Moon and Why It Changes	`days/day-01-moon-and-why.md`	Torch-and-ball demo
2	The Two Pakṣas	`days/day-02-paksha.md`	Diagram-build, exit ticket
3	Naming the 15 Tithis	`days/day-03-fifteen-tithis.md`	Tithi-ordinal drill; formative quiz
4	Build a Tithi Calendar	`days/day-04-tithi-calendar.md`	`activities/activity-01-build-tithi-calendar.md`
5	Lunar Months and Festivals	`days/day-05-lunar-months.md`	`quizzes/formative.md` (Part B)
6	The Pañcāṅga — Five Limbs	`days/day-06-panchanga.md`	Compare-contrast worksheet
7	Eclipses — Why and When	`days/day-07-eclipses.md`	`activities/activity-02-eclipse-geometry.md`
8	Project Session 1: Plan	`days/day-08-project-session-1.md`	Project draft submitted

Day	Session title	Lesson file	Activity / assessment
9	Cross-cultural Lunar Calendars + Project Build	`days/day-09-cross-cultural.md`	Project draft submitted
10	Final Assessment + Showcase	`days/day-10-final-assessment.md`	`quizzes/summative.md` + project

Pacing notes

- Days 1–3 are content-heavy. If your class needs a half-day buffer, split Day 3 across two sessions and roll Days 5–10 forward by one slot. Drop Day 9's cross-cultural segment if you must compress; assign it as reading.
- Day 4's tithi calendar benefits from advance setup — print one *pañcāṅga* page (or open an astronomy app) for the current month before class begins.
- The 10-night Moon observation diary (homework) starts on Day 1 and runs across the module. Remind students every day for the first three days.
- Project work (Days 8–9) assumes you've decided pair vs. individual by Day 5.

Suggested running theme

Open each class with the same question on the board: **"Did you see the Moon last night? What shape was it?"** Two students share for 30 seconds each. By Day 10 every student will have spoken at least twice. This builds the observation habit the module is designed to grow.

Formative vs. summative weight

- Formative quizzes (Days 3 and 5) are **not graded** — they are diagnostic. Use them to decide who needs scaffolding on Day 6.
- Summative quiz (Day 10) — see rubric in `assessments/rubric.md`. Combined with project, total 40 marks.

What students leave the module with

- A 10-night Moon-shape diary in their notebook with tithi names.
- A completed tithi calendar for one calendar month.
- The 15 tithi names with English translation, in their own handwriting.
- One completed project artefact.
- The habit of looking up — and noticing the Moon — most nights.

Homework

Homework — Module 6 Moon Phases & Tithi (Middle)

Homework tasks per day. None should take more than 25 minutes for a Middle-band student. Skip a day if it conflicts with another subject's deadline.

The **Moon diary** is a running task: every night from Day 1 to Day 10, sketch the Moon's shape, note the date/time/cloud cover, and (from Day 3 onwards) name the tithi.

Day 1 → Day 2

Task: Start your **10-night Moon diary**. Tonight: look at the Moon. Sketch its shape in your workbook. Note the date, time, and "clear / partly cloudy / overcast." If overcast, look up the phase on a Moon-phase app or NASA's published phase calendar.

Time: 5 min (looking) + 5 min (sketching).

Day 2 → Day 3

Task: Continue the Moon diary. Tonight, add **which pakṣa you are in** to your sketch.

Bonus: Find out from a parent or a calendar — has anyone in your family observed *amāvasyā* or *pūrṇimā* this month for any reason (fast, ritual, prayer)?

Time: 10 min.

Day 3 → Day 4

Task: Memorise the 15 tithi names (śukla pakṣa). Quiz yourself or be quizzed at home. **Be ready to recite all 15 in under 60 seconds at the start of tomorrow's class.**

Continue the Moon diary — now naming the tithi as well as the pakṣa.

Time: 15 min.

Day 4 → Day 5

Task: Compare last night's Moon-diary sketch with the tithi calendar you built in class. Does the shape match the predicted shape for the tithi?

Also: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Time: 10 min.

Day 5 → Day 6

Task: Submit your **project plan paragraph** in writing.

Format: one paragraph, 5–8 sentences, stating: which lunar phenomenon, which format (poster / demo / story / video / infographic / slide deck), and what audience will see/hear/feel.

Time: 15 min.

Day 6 → Day 7

Task: Read NASA's published lunar phase page for the current year (search "NASA Moon phases [year]"). Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Continue the Moon diary.

Time: 20 min.

Day 7 → Day 8

Task: Bring ALL project materials to tomorrow's class. The build session starts immediately. Anything you forget cannot be retrieved during class time.

Bonus: Look up the date of the next solar eclipse and the next lunar eclipse visible from your location. Note in your workbook.

Time: 15 min.

Day 8 → Day 9

Task: Finish at home what you couldn't finish in class today. Tomorrow includes the cross-cultural lesson + refining, not starting.

Time: variable — but cap at 30 min, then stop and rest. Done-is-better-than-perfect.

Day 9 → Day 10

Task: Final touches. Make sure you have: - [ ] The artefact (poster / video file / printed story / etc.) - [ ] Your script or notes - [ ] Your completed Moon diary (10 nights) - [ ] A working version of your 3-minute presentation in your head

Time: 15 min.

After Day 10

No homework. Module is complete. Take a breath. (If your next module begins immediately, check that module's README for prerequisite reading.)

Optional follow-up: Keep the Moon diary going for another 20 days. By the end you'll have observed (or noted from a calendar) a full month of phases — and you'll know exactly which night to look for the next *pūrṇimā*.